Who pays for a carbon tax?

ECON-4210
Economics of Climate Change
Class 20

Prof. Casey Wichman

Georgia Institute of Technology

Today's class

1. Questions?
2. Attendance
3. Housekeeping
    1. Problem set 3 due Friday, November 13
    2. Wednesday: Exam 3 review and Ghost Emissions round 8
    3. Exam 3 on Monday, November 16 (classes 15–20)
4. Who pays for a carbon tax?
5. For next class…

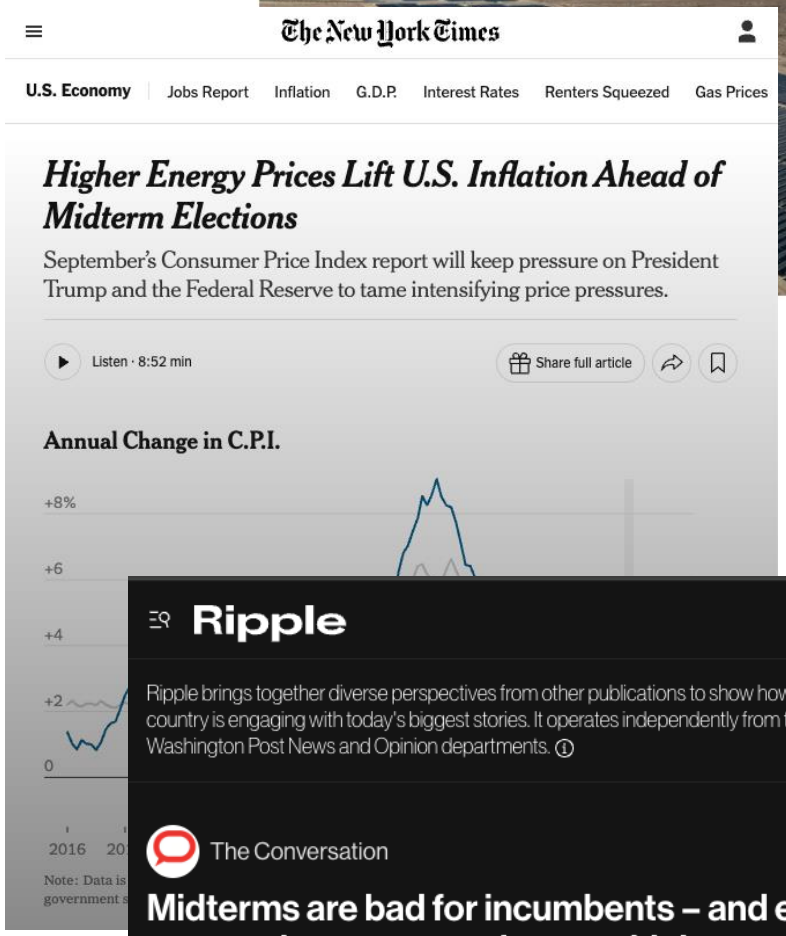
Two questions about a carbon tax

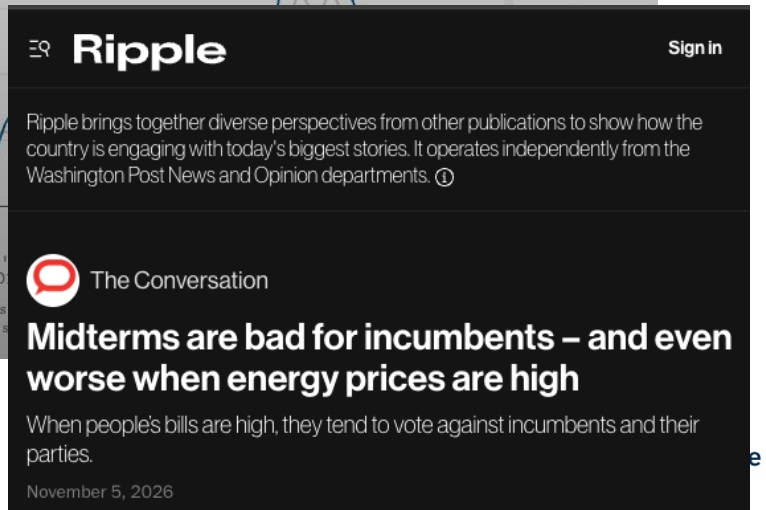
- In our policy experiment, a $61 tax collected $5,856 from our eight firms.
- **Who actually bears that cost?**
  - The firms that send the check to the government, their customers, their workers, or their owners?
- **What happens to the $5,856?**
  - The answer to the first question depends on the answer to this one.
- Economists like carbon taxes because they reduce emissions at least cost.
  - That's an efficiency result
  - Who pays the tax matters for concerns about distributional equity (and political feasibility…)

What is a regressive tax?

- A tax is **regressive** if low-income households pay a larger % of income than high-income households.
  - Progressive: the opposite.
  - Proportional: the same share for everyone.
- Example: every household pays a $500 tax per year.
  - $500 is 2.0% of income for a household earning $25,000.
  - $500 is 0.5% of income for a household earning $100,000.
  - → this tax is regressive
- Low-income households spend a larger share of their budgets on energy. So a tax on energy looks regressive at first glance.
- Gasoline, electricity prices were a big contributing factor for recent midterm elections.





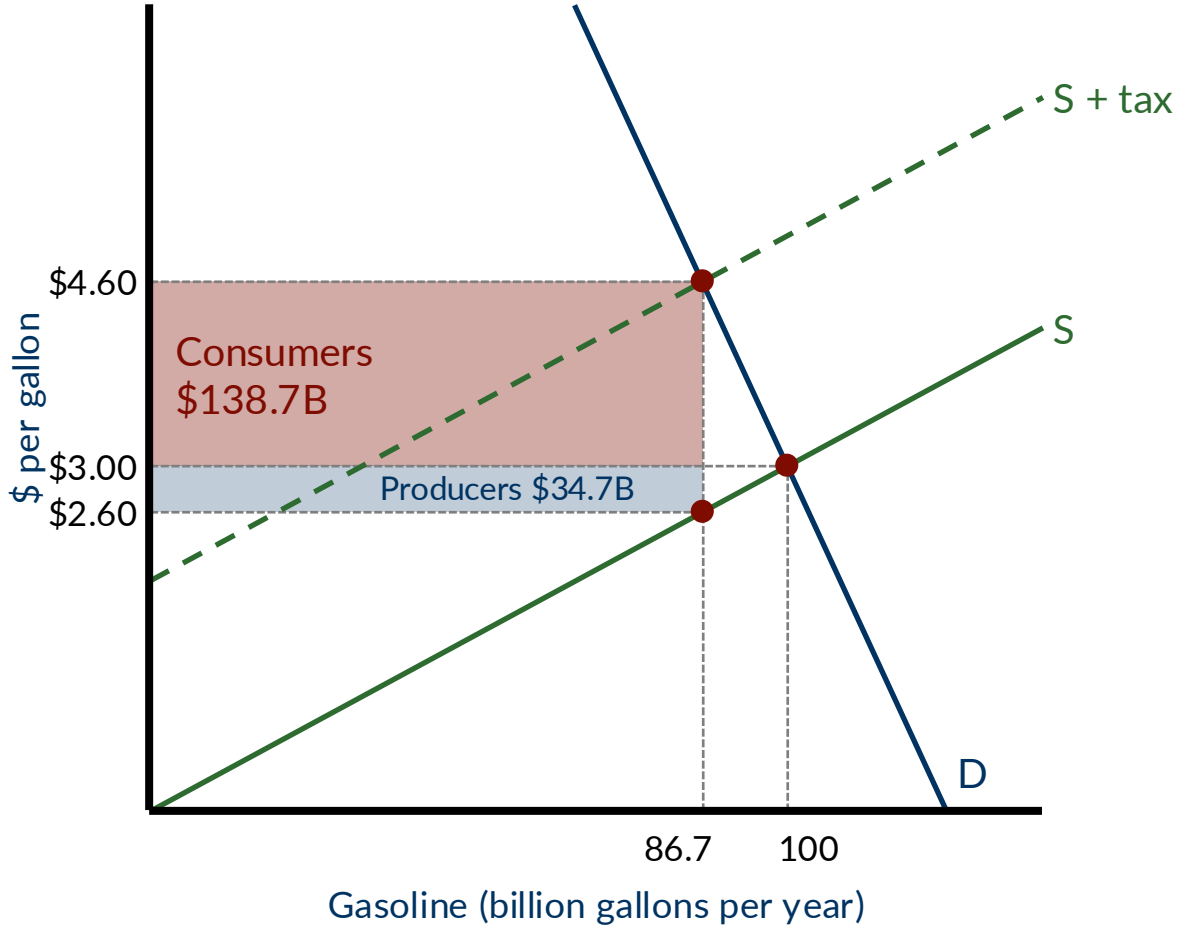


Who sends the check vs. who pays

- ***Statutory incidence:*** who is legally required to pay the tax to the government.
  - For a carbon tax, usually fuel suppliers and power plants.
- ***Economic incidence:*** whose income actually decreases.
  - Prices adjust and the burden can be "passed through" to consumers (higher prices) or back to workers and owners (lower wages and profits).
- In a competitive market, *it doesn't matter who writes the check to the government*. A $2 tax collected from gas stations or from drivers leads to the same economic incidence.
- What matters: how responsive buyers and sellers are to price changes (i.e., the elasticities of supply and demand).

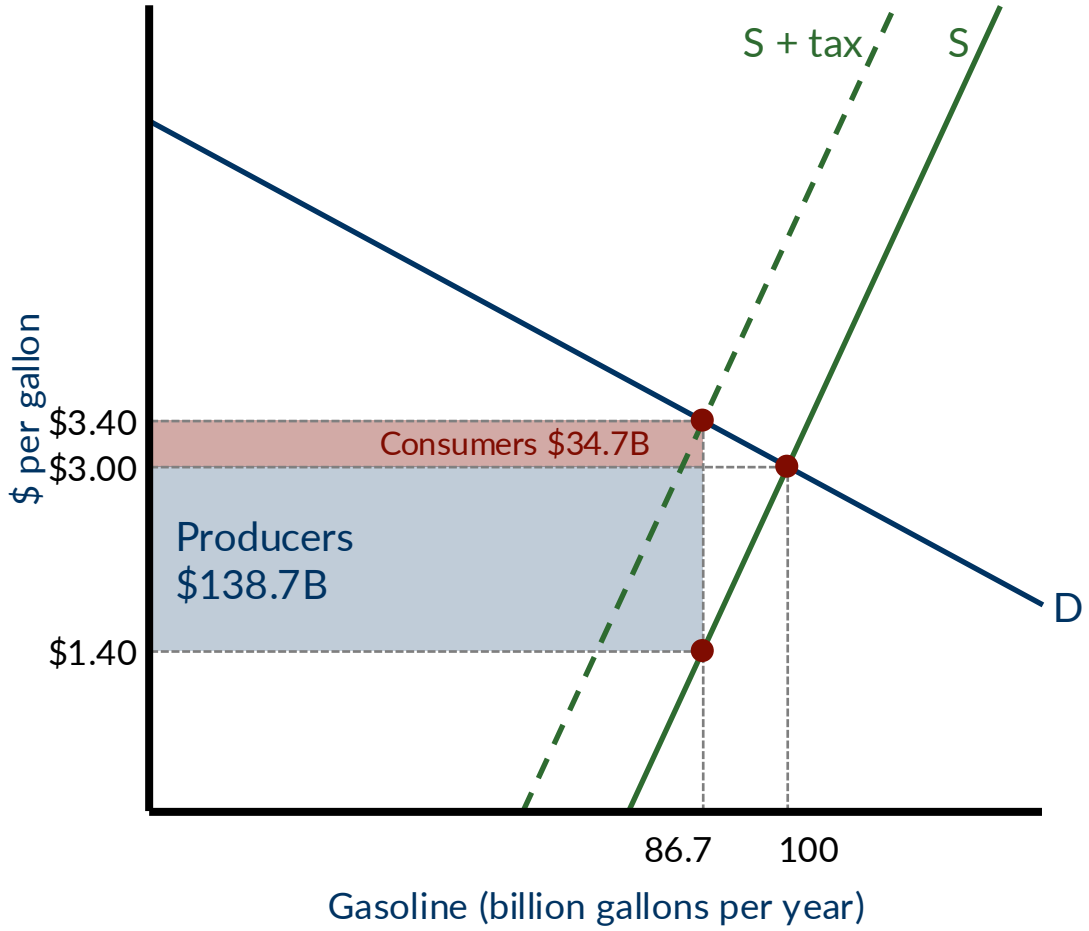
A $2 per gallon tax on gasoline

- Before the tax:
  - 100 billion gallons at $3.00.
- The tax shifts supply up by $2.
- Demand is inelastic ($\varepsilon_D = -0.25$). Supply is more elastic ($\varepsilon_S = 1$).
- **Consumers pay $4.60. Producers keep $2.60.**
  - Consumers bear $1.60 of the $2 tax (80%).
  - Why 80%? $|\varepsilon_D| \cdot x = \varepsilon_S \cdot y$, where x and y are the % price changes for consumers and producers.
  - Consumers' share = $x/(x + y) = \varepsilon_S/(\varepsilon_S + |\varepsilon_D|) = 0.8$
  - In total, consumers pay $1.60 × 86.7 = $138.7 billion.
  - Producers pay $0.40 × 86.7 = $34.7 billion.
- Revenue: $2 × 86.7 = $173 billion.



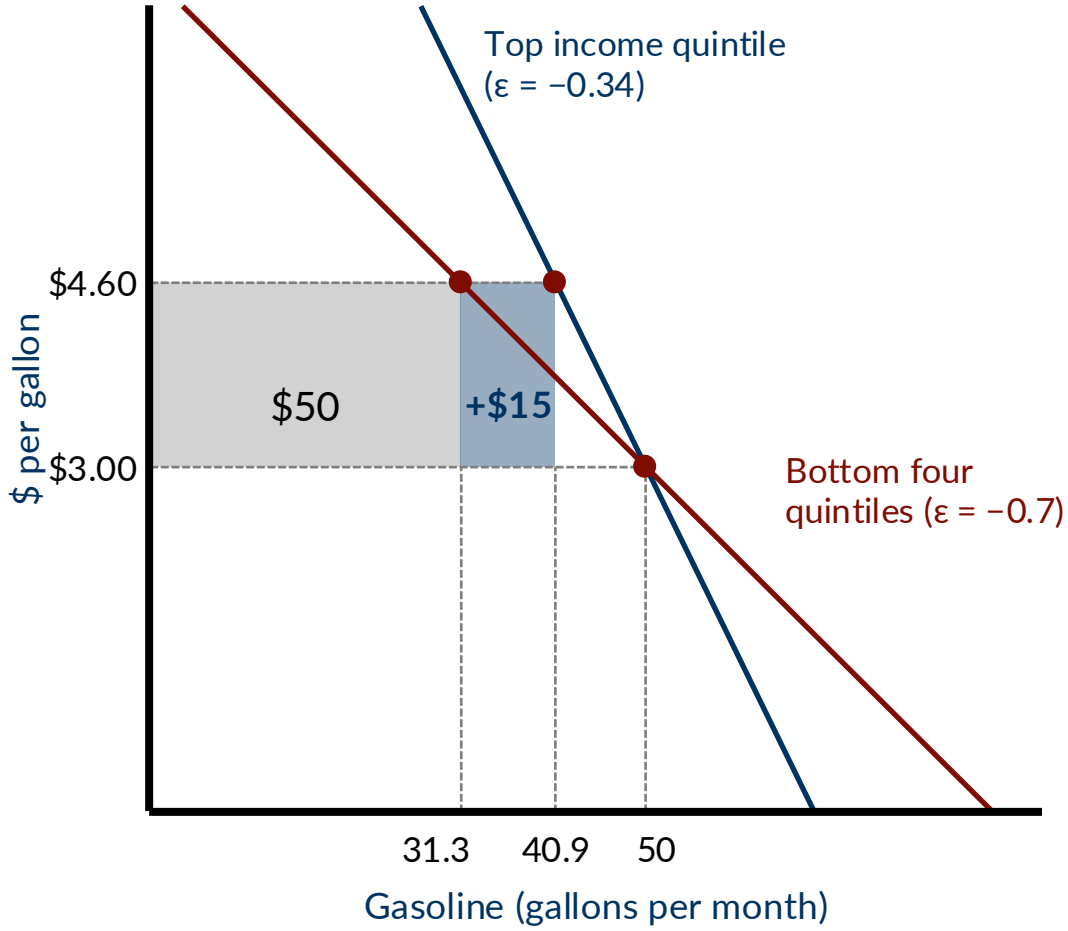
Same tax, inelastic supply

- Now flip the elasticities: $\varepsilon_D = -1$ and $\varepsilon_S = 0.25$.
- **Consumers pay $3.40. Producers keep $1.40.**
  - Producers bear $1.60 of the $2 tax (80%).
  - In total, producers pay $1.60 × 86.7 = $138.7 billion.
  - Consumers pay $0.40 × 86.7 = $34.7 billion.
- Same tax, same quantity, same $173 billion in revenue.
- The burden is reversed.



Among consumers, the slope of demand matters

- Every driver pays the same $1.60 more per gallon. But, tax burden depends on elasticity of demand.
- Two households each buy 50 gallons a month at $3.00.
  - Top income quintile (ε = −0.34): buys 40.9 gallons, loses $73 a month.
  - Bottom four quintiles (ε = −0.7): buys 31.3 gallons, loses $65 a month.
- **The household with more elastic demand avoids more of the tax.**
  - Gray: both pay $1.60 × 31.3 = $50.
  - Blue: the top quintile pays $15 more.
- Ignoring behavior ($1.60 × 50 = $80 each) overstates the burden of a gas tax and how regressive it is (West and Williams, 2004).



West, S.E., and R.C. Williams III (2004). Estimates from a consumer demand system: Implications for the incidence of environmental taxes. Journal of Environmental Economics and Management 47(3): 535–558. Elasticities from the working paper version, Table 3. nber.org/papers/w9152

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The less elastic curve pays more of the tax

- **Share of the tax borne by consumers = ε_S / (ε_S + $|\varepsilon_D|$)**
  - Gasoline example: 1 / (1 + 0.25) = 0.8
- Gasoline demand is inelastic in the short run. Most people have no quick substitute for driving to work.
- Oil trades on a world market, so supply to any one country is elastic.
  - Consumers bear most of a fuel tax.
- Coal mines and oil wells can't relocate. Where supply is inelastic, more of the tax falls on the owners and workers in those industries.
- Both sides are more elastic in the long run (e.g., new cars, new power plants).

Two ways a carbon tax reaches households

- **Use side: the prices of goods that the households buy increase.**
  - Low-income households spend a larger share of their budgets on energy.
  - → This side is regressive.

- **Source side: household incomes change.**
  - Wages and returns to capital in carbon-intensive industries decrease.
  - Many transfer payments (e.g., Social Security) are indexed to inflation, so they increase with prices.
  - Low-income households get a larger share of income from transfers.
  - → This side tends to be progressive.

- In a model of the US economy, the progressive sources side largely offsets the regressive uses side (Goulder et al., 2019).

Goulder, L.H., M.A.C. Hafstead, G. Kim, and X. Long (2019). Impacts of a carbon tax across US household income groups: What are the equity-efficiency trade-offs? Journal of Public Economics 175: 44–64. doi.org/10.1016/j.jpubeco.2019.04.002

Direct vs. indirect effects

- **Direct:** a carbon tax increases the price of gasoline, electricity, and natural gas.
- **Indirect:** it also increases the price of anything made or moved with energy (e.g., airfare, groceries, your Uber).
- Low-income households spend more of their budgets on energy directly. High-income households spend more on goods with indirect effects (e.g., air travel, recreation).
  - Counting indirect effects makes an energy tax look less regressive.
- Indirect effects are large. For a proposed US energy tax in the 1990s, more than half of the average household's burden was indirect (Bull, Hassett, and Metcalf, 1994).

Bull, N., K.A. Hassett, and G.E. Metcalf (1994). Who pays broad-based energy taxes? Computing lifetime and regional incidence. The Energy Journal 15(3): 145–164. doi.org/10.5547/ISSN0195-6574-EJ-Vol15-No3-8

Are energy expenditures regressive?

- Energy spending as a share of total spending, by spending decile (1 = poorest).
- Taxes increase energy expenditures proportionally.
- Electricity: the share decreases with income in the US, the UK, and Mexico.
  - A tax on electricity is regressive everywhere.
- Motor fuels: slightly decreasing in the US, hump-shaped in the UK, increasing in Mexico.
  - Many of the poorest households in Mexico don't own a car. A fuel tax there is progressive.



Figure 2 Average household electricity and motor fuels expenditures as percentage of total spending for the United States, the UK and Mexico, by expenditure decile.
Notes: Decile 1 is poorest, 10 is richest.
Sources: Instituto Nacional de Estadística y Geografía (2012), Office of National Statistics (2013), and Bureau of Labor Statistics (2014).

Pizer, W.A., and S. Sexton (2019). The distributional impacts of energy taxes. Review of Environmental Economics and Policy 13(1): 104–123. doi.org/10.1093/reep/rey021

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Dividends can make taxes less regressive

A stylized carbon tax with all revenue returned as a dividend of $1,100 per household.

Income quintile	Income	Carbon tax paid	Tax as % of income	Dividend	Net gain	Net as % of income
1 (poorest)	$20,000	$500	2.5%	$1,100	+$600	+3.0%
2	$45,000	$750	1.7%	$1,100	+$350	+0.8%
3	$75,000	$1,000	1.3%	$1,100	+$100	+0.1%
4	$115,000	$1,250	1.1%	$1,100	−$150	−0.1%
5 (richest)	$250,000	$2,000	0.8%	$1,100	−$900	−0.4%
Average	**$101,000**	**$1,100**		**$1,100**	**$0**	

- Before the dividend, the tax is regressive: 2.5% of income for the poorest quintile, 0.8% for the richest.
- **After the dividend, the bottom three quintiles come out ahead.**
- High-income households pay more dollars of tax because they consume more carbon, directly and indirectly (Grainger and Kolstad, 2010).

Grainger, C.A., and C.D. Kolstad (2010). Who pays a price on carbon? Environmental and Resource Economics 46: 359 –376. doi.org/10.1007/s10640-010-9345-x

What should we do with the revenue?

- Return it as an equal dividend to every household.
  - Most progressive. Most low- and middle-income households come out ahead.
- Cut labor income taxes.
  - Burden is roughly proportional across income groups.
- Cut capital income taxes.
  - Smallest total cost to the economy, but the gains go mostly to high-income households.
- The most progressive use of revenue is not the cheapest one. There is a trade-off between equity and efficiency (Williams et al., 2015; Goulder et al., 2019).
- Our firms faced the same question. An equal rebate of the $5,856 made every firm better off than under the standard.

Williams, R.C. III, H. Gordon, D. Burtraw, J.C. Carbone, and R.D. Morgenstern (2015). The initial incidence of a carbon tax across income groups. National Tax Journal 68(1): 195–214. doi.org/10.17310/ntj.2015.1.09
Goulder, L.H., M.A.C. Hafstead, G. Kim, and X. Long (2019). Impacts of a carbon tax across US household income groups: What are the equity-efficiency trade-offs? Journal of Public Economics 175: 44–64. doi.org/10.1016/j.jpubeco.2019.04.002

Energy use differs *within* income groups too

Two GT professors in the middle quintile. Both receive the $1,100 dividend.

Household	Home	Getting to work	Carbon tax paid	Dividend	Net gain
A	Large house in Roswell	30-mile commute in an SUV	$1,700	$1,100	−$600
B	Condo in Midtown	Walks to work	$300	$1,100	+$800
Average			**$1,000**	**$1,100**	**+$100**

- **The quintile average says these households gain $100. Neither household is average.**
- Vertical equity compares households across income groups.
- Horizontal equity compares households within an income group.
- Differences within income groups are about as large as differences across them (Pizer and Sexton, 2019).

Horizontal equity

- Within an income group, energy use varies with housing, climate, commutes, and family size.
  - Among the poorest 10% of US households, one-third spent nothing on gasoline. One-sixth spent more than 10% of their budgets on it (Poterba, 1991).
  - For electricity, the spread within income groups is about as large as the spread across them (Pizer and Sexton, 2019).

- Fixes are difficult.
  - Compensate by energy use? Then the rebate cancels the tax. A household that drives more gets more back.
  - Compensate by place? Canada paid a 20% rural top-up on its carbon rebate.

- Why care? Several tax-and-rebate designs are progressive on average but impose small losses on most low-income families (Cronin, Fullerton, and Sexton, 2019). The "losers" can make the policy difficult to sell politically.

Poterba, J.M. (1991). Is the gasoline tax regressive? Tax Policy and the Economy 5. doi.org/10.1086/tpe.5.20061803
Pizer, W.A., and S. Sexton (2019). The distributional impacts of energy taxes. Review of Environmental Economics and Policy 13(1): 104–123. doi.org/10.1093/reep/rey021
Cronin, J.A., D. Fullerton, and S. Sexton (2019). Vertical and horizontal redistributions from a carbon tax and rebate. Journal of the Association of Environmental and Resource Economists 6(S1): S169–S208. doi.org/10.1086/701191

Perceived burden matters too

- France, 2018: planned increases in fuel taxes set off the ”yellow vest” protests, mostly among drivers outside big cities.
  - The government suspended the increase in December 2018.

- Canada, 2019–2025: a federal fuel charge, with revenue returned to households as quarterly rebates.
  - Most households received more than they paid.
  - In surveys, many didn’t know they received a rebate, and most underestimated its size (Mildenberger et al., 2022).
  - Canada ended the consumer fuel charge on April 1, 2025.

- The tax is very salient at the pump, but the rebate is less salient (arrives quarterly).

Mildenberger, M., E. Lachapelle, K. Harrison, and I. Stadelmann-Steffen (2022). Limited impacts of carbon tax rebate programmes on public support for carbon pricing. Nature Climate Change 12(2): 141–147. doi.org/10.1038/s41558-021-01268-3
Department of Finance Canada (2025). Removing the consumer carbon price, effective April 1, 2025. canada.ca/...

Key takeaways

- The firm that sends the tax revenue check to the government isn't necessarily the one that pays.
  - The less elastic side of the market bears more of the tax.
- On the uses side, energy taxes look regressive, especially for electricity. Indirect effects and the sources side make them less so.
- What we do with the revenue decides the final answer. An equal dividend can make a carbon tax progressive on average.
- A dividend fixes differences across income groups, not within them.
- Income is not the only dimension. Burdens also differ by region, race, and age.

Next class…

- Problem set 3 due Friday, November 13.
- Next class: Exam 3 review and Ghost Emissions round 8. Bring questions.
- Exam 3 is Monday, November 16, and covers classes 15–20.

www.caseyjwichman.com/econ4210#week12